

Math 421: Homework 1

Solutions

Due: Sept 12, 2025

Problem 1: Determine whether the following statements are true or false and provide a short justification (can be an example or a counterexample where appropriate) -

- A. For every integer x , $x \leq 5$ and $x > 3$.
FALSE. Let $x = 7$.
- B. There exists an integer x such that $x \leq 5$ and $x > 3$.
TRUE. Let $x = 4$.
- C. There exists a unique integer x such that $x \leq 5$ and $x > 3$.
FALSE. x can be 4 or 5.
- D. There exists an integer x such that for all integers y , $xy = x$.
TRUE. Let $x = 0$.
- E. For all integers x there exists an integer y such that $xy = x$.
TRUE. For every x , let $y = 1$.

- Problem 2:**
- A. If a function, f , is differentiable, then f is continuous.
 p : “ f is differentiable”
 q : “ f is continuous”
 - B. Converse: If a function f , is continuous, then f is differentiable.
 - C. Contrapositive: If a function f is not continuous, then f is not differentiable.

Problem 3: Use a truth table to show that the converse of an “if..., then...” statement **not** equivalent to the original statement

p	q	$p \Rightarrow q$	$q \Rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

As the last two columns do not have the same truth values, the two statements are not equivalent.

Problem 4: If x and y are odd integers, then xy is an odd integer.

Proof: Let x and y be odd integers. That is, there exists integers k and l such that $x = 2k + 1$ and $y = 2l + 1$. Then

$$xy = (2k + 1)(2l + 1) = 4kl + 2k + 2l + 1 = 2(2kl + k + l) + 1.$$

By the closure of the integers under addition and multiplication, $2kl + k + l$ is an integer. So xy satisfies the definition of an odd integer.

Problem 5: Suppose x and y are two integers with the same parity (that is, either x and y are both even or x and y are both odd), then $x + y$ is even.

Case 1: Let x and y be two even integers. Then there exists integers k and l such that $x = 2k$ and $y = 2l$. Thus

$$x + y = 2k + 2l = 2(k + l).$$

By the closure of the integers under addition, $k + l$ is an integer, so $x + y$ satisfies the definition of an even integer.

Case 2: Let x and y be two odd integers. Then there exists integers m and n such that $x = 2m + 1$ and $y = 2n + 1$. Thus

$$x + y = 2m + 1 + 2n + 1 = 2m + 2n + 2 = 2(m + n + 1).$$

By the closure of the integers under addition, $m + n + 1$ is an integer, so $x + y$ satisfies the definition of an even integer.

Problem 6: For all real numbers x and y , $(x + y)^2 = x^2 + y^2$ if and only if $x = 0$ or $y = 0$.

$\Rightarrow$: Let x and y be real numbers such that $(x + y)^2 = x^2 + y^2$. We have

$$(x + y)^2 = x^2 + 2xy + y^2 = x^2 + y^2.$$

Thus $2xy = 0$, and either $x = 0$ or $y = 0$.

$\Leftarrow$: Without loss of generality, let $x = 0$ (in the case $y = 0$, we can interchange the names of the variables and the proof still holds). We have

$$x^2 + y^2 = 0^2 + y^2 = y^2 = (0 + y)^2 = (x + y)^2.$$

Thus, $(x + y)^2 = x^2 + y^2$.